

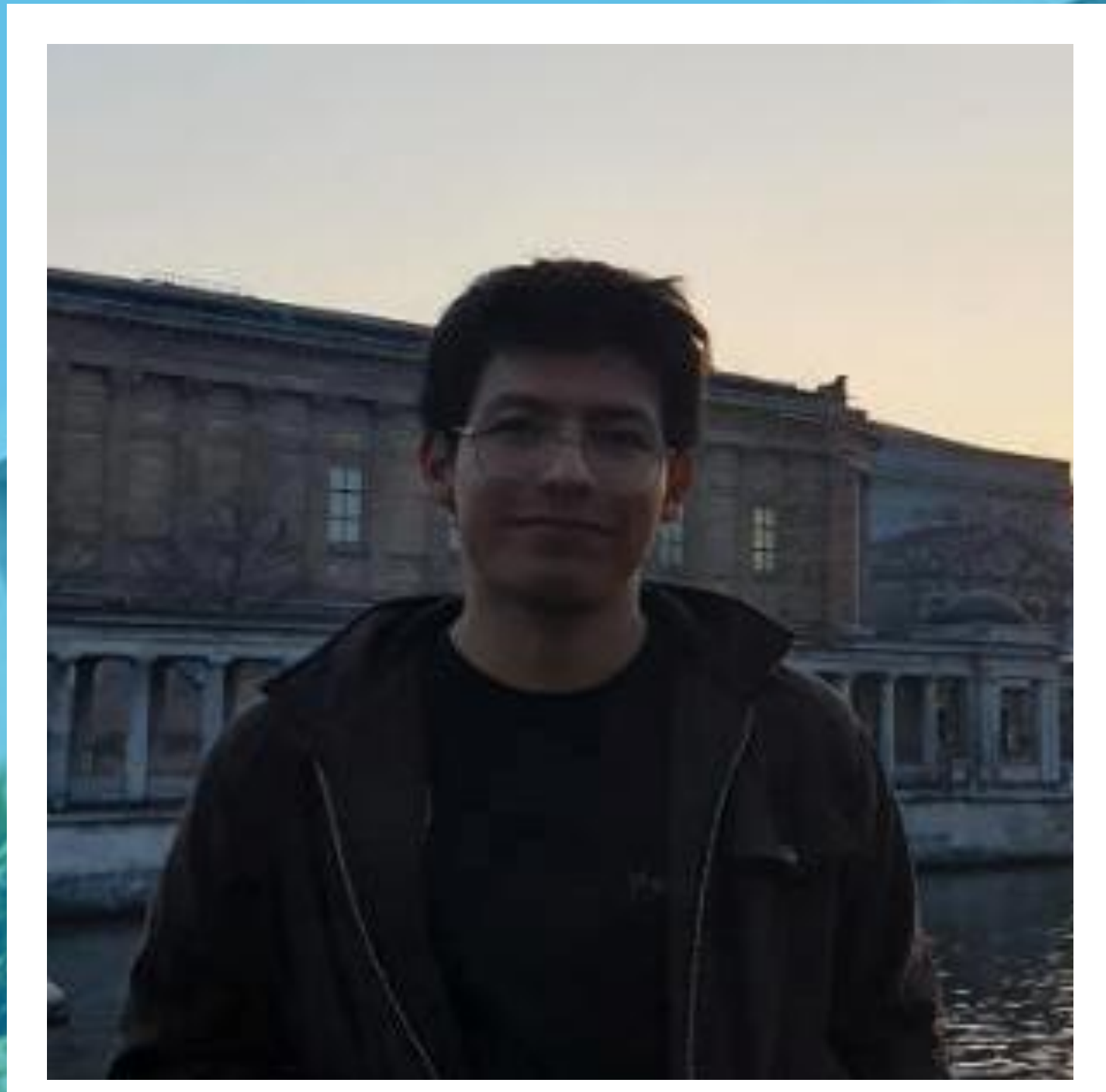
1. Introduction

Internet of Things

Executive Summary

2

- **Motivation:** IoT emerges as a solution to define smart devices as “things”.
- **Problem:** We need to define a scope of IoT study and the main design goals and platforms.
- **Overview:**
 - CS5055 course logistics.
 - Internet of Things (IoT) introduction.
 - Review of IoT fundamentals and computing systems.
- **Conclusion:** IoT emerges as an integral platform to propose next-gen designs for providing services using networking and “things”.



Msc. Carlos Williams Valencia
cwilliams@utec.edu.pe
Lecturer
Department of Computer Science

He has a master's degree in Nanophysics from the Freie Universität Berlin (Germany) and the École Polytechnique (France). He has completed internships in different institutions and companies (INICTEL-UNI, STMicroelectronics and EPFL) where he has obtained extensive experience in semiconductor physics, optics and device design. His current research interests focus on photonics, design optimization and nanophysics.

Outline

4

Course Logistics

Introduction

IoT design perspective

Basic Circuits

Conclusions

CS5055 Internet of Things

5

- Introduces fundamental concepts of IoT paradigm for design and implementation.
- **Objectives:**
 - Show the principles of interconnected IoT systems.
 - Explain the interaction between computing and sensing for IoT.
 - Implement an IoT application system as a solution.
- **Content:**

Distributed in two modules:

 - First module, from Week 1 to Week 8
 - Second module, from Week 9 to Week 16

Evaluation

6

- For more details, please refer to the course Syllabi in Canvas or:

Teoría 45%
Examen Parcial (EP) 20%
Examen Final (EF) 25%
Práctica y/o laboratorio 55%
Laboratorio 1 (L1) 10%
Laboratorio 2 (L2) 10%
Laboratorio 3 (L3) 10%
Laboratorio 4 (L4) 10%
Proyecto (P1) 05%
Proyecto (P2) 10%
<i>Nota: P2 no puede ser presentado si no se ha presentado el P1</i>
100%

Proyecto se va a empezar a discutir desde las primeras semanas en la sesión de laboratorio.

La primera nota será el P1 que consiste en una presentación de la idea de proyecto, necesaria para poder continuar con el P2.

El P2 se evaluará en formato de una feria de proyecto la última semana de clases.

Los primeros 2 puestos de la feria recibirán automáticamente una bonificación en sus exámenes finales.

Evaluation

- For more details, please refer to the course Syllabi in Canvas or:

Teoría 45%

Examen Parcial (EP) 20%

Examen Final (EF) 25%

Práctica y/o laboratorio 55%

Laboratorio 1 (L1) 10%

Laboratorio 2 (L2) 10%

Laboratorio 3 (L3) 10%

Laboratorio 4 (L4) 10%

Proyecto (P1) 05%

Proyecto (P2) 10%

*Nota: P2 no puede ser presentado
si no se ha presentado el P1*

100%

Las notas de laboratorio son la suma del informe (12 pts) + Test de salida (8 pts) :

Laboratorio 1 (Test Salida + Entrega de informe) = Martes/Miercoles- Semana 5

Laboratorio 2 (Test Salida + Entrega de informe) = Miercoles/Viernes- Semana 7

Laboratorio 3 (Test Salida + Entrega de informe) = Martes/Miercoles- Semana 11

Laboratorio 4 (Test Salida + Entrega de informe) = Martes/Miercoles- Semana 13

Important Rules

Do not:

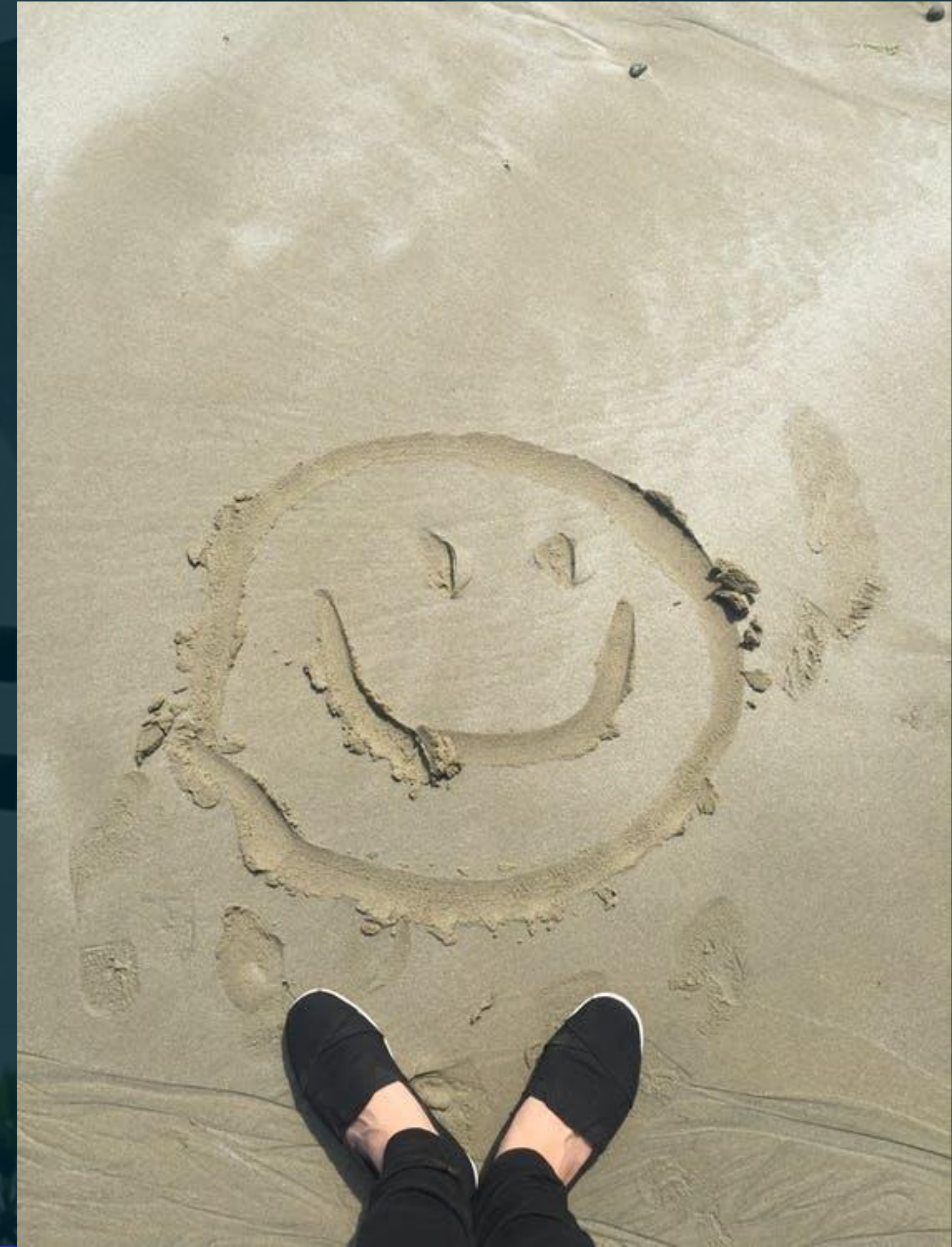
- **Publish your solution repos or share with other students before evaluation in canvas.**
- **Use partial or entire solutions and code implementations** from: a) online repositories, c) or other students (including those who have already taken the course).
- **In doubt, ask instructor**

Do:

- Discuss ideas and problems with other students
- Ask your instructor

- **All course communication via Piazza. Use email for emergencies.**
 - Do not send direct messages, use the server channel. Do not be afraid to ask 😊
 - **Use email for emergencies.**
 - **IoT** forum: do not send code, solutions, or illegal items.
 - Piazza
 - Laboratory professor: Gabriel Arias
 - Martes: 18:00 – 20:00/ Miércoles 20:00 – 22:00
 - Miércoles: 18:00 – 20:00 / Viernes 20:00 – 22:00

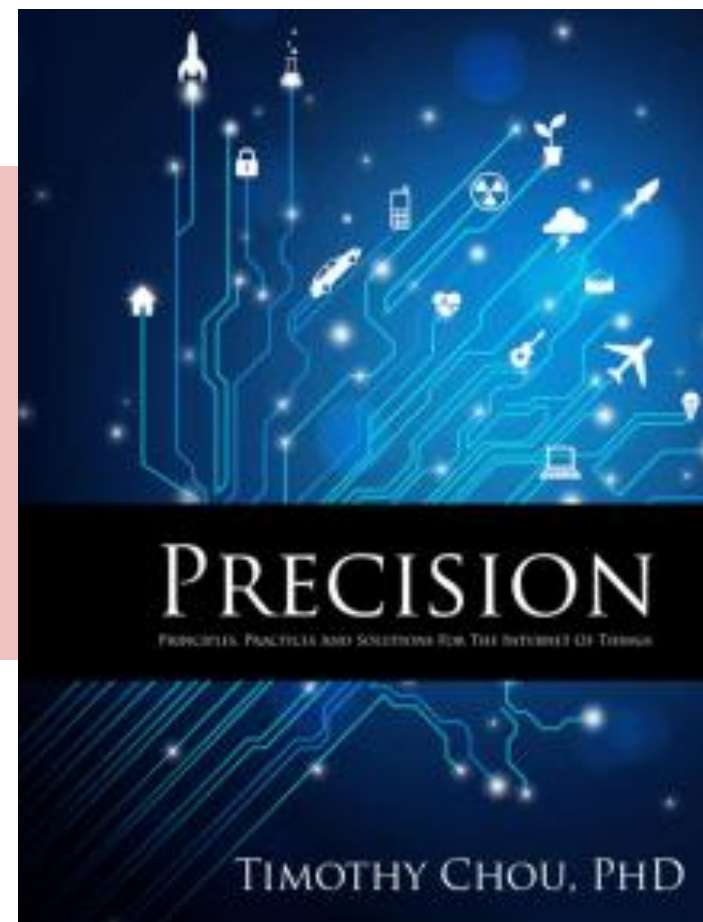
1 Enjoy the IoT trip 😊



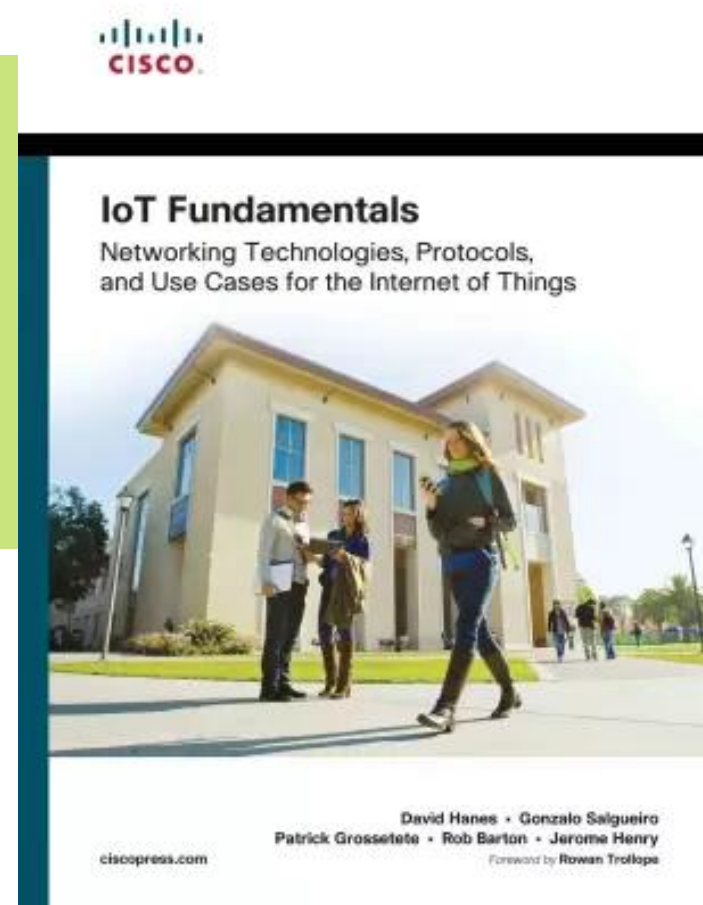
Books

12

- The **books** for our IoT course are:



Chou T. (2020). Precision: Principles, Practices and Solutions for the Internet of Things. Lulu Press.



Hanes, D., Salgueiro, G., Grossetete, P., Henry, J. y Barton, R. (2017). IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things. Cisco Press.

Outline

13

Course Logistics

Introduction

IoT design perspective

Basic Circuits

Conclusions

What is expected from IoT?

14

<https://www.youtube.com/watch?v=mkIV-DnKgwg>



#Arm #IoT #InternetofThings

IoT on Arm: It's Time to Build

“Fun” Internet-connected devices

15

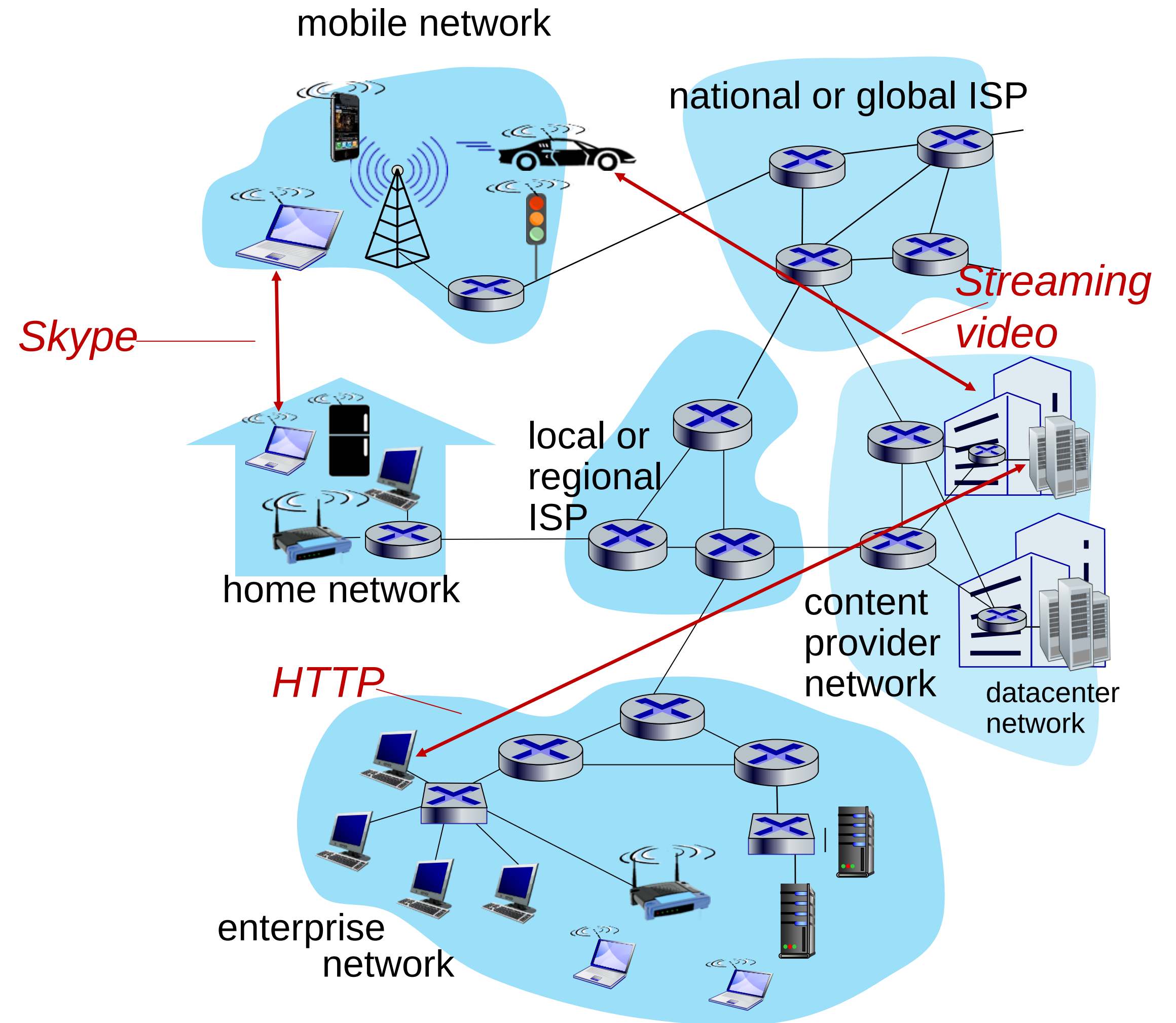


Others?

Recall: the Internet: a “services” view

16

- **Infrastructure** that provides services to applications:
 - Web, teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides **programming interface** to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service



What about “Things”?

17

- **Set of entities:** smart devices, sensors, human beings, and any other objects aware of its context and is able to communicate with other entities
 1. Accessible at any time, anywhere.
 2. Integration of mobile devices, edge devices like routers and smart hubs, and humans in the loop as controllers.
- **Need to support** a diverse set of **devices** and **communication protocols**.
- **Example:** 1) from tiny sensors, to 2) also powerful back-end servers that are utilized for data analysis



Internet of Things (IoT)

18

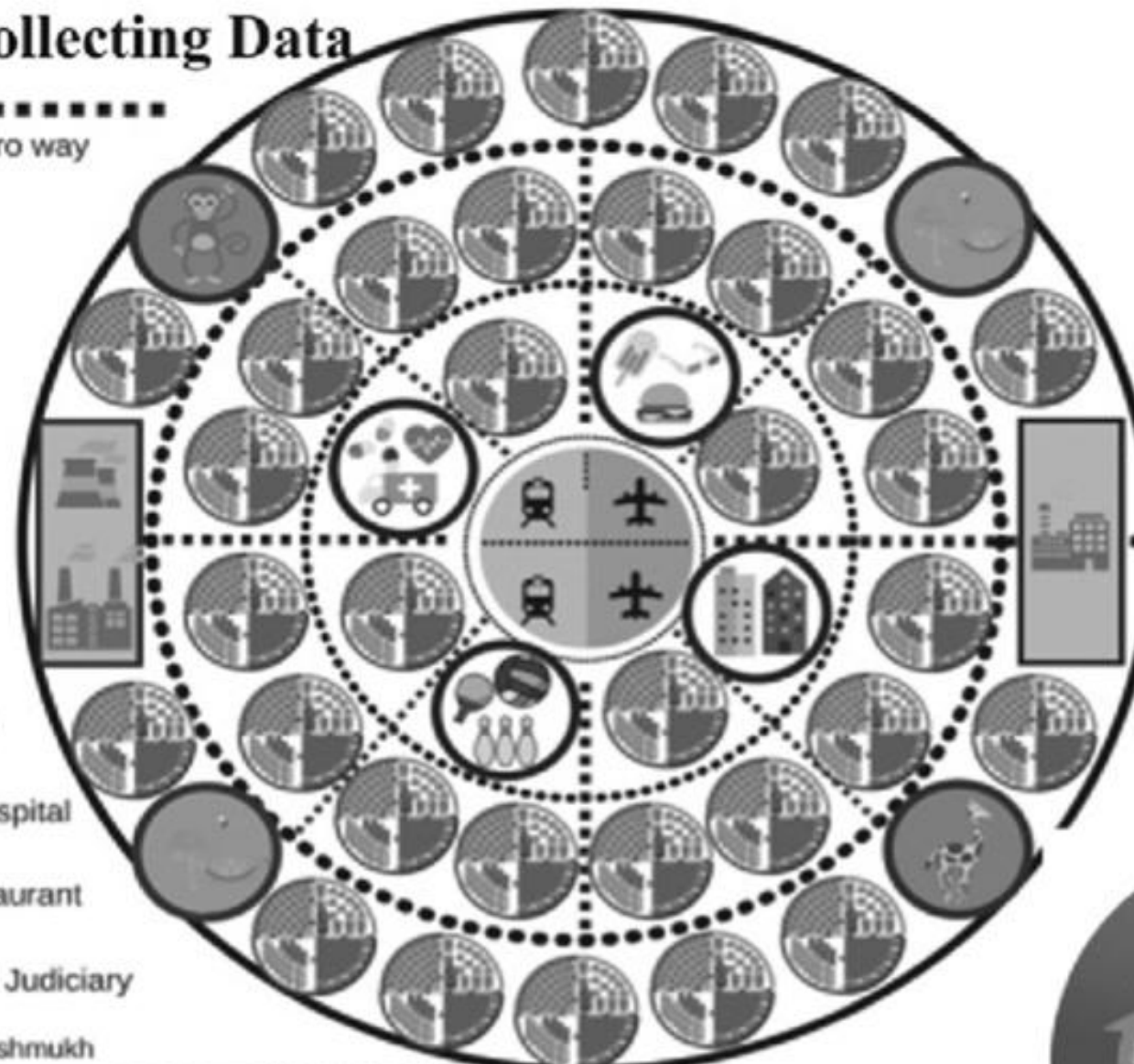
. IoT: Internet + Things

- . Kevin Ashton is accredited for using the term “Internet of Things”, 1999 - on supply-chain management conference.
- . **Connectivity** (also ‘smartness’) is a fundamental aspect for IoT:
 - . **Initially:** Radio-Frequency Identification (RFID) used to be the dominant technology wireless sensor networks (WSN)
 - . **Now:** Bluetooth-enabled devices augmented the mainstream adoption of the IoT trend.

· Sensing

Sensing and Collecting Data

- Road + Super speed metro way
 - A Single Block
 - Water Body
 - Green Cover
 - Factory
 - Railway Station
 - Airport
 - Recreation Center
 - Multi Speciality Hospital
 - Multiplex and Restaurant
 - Administration and Judiciary
- Designed by Jayati Deshmukh

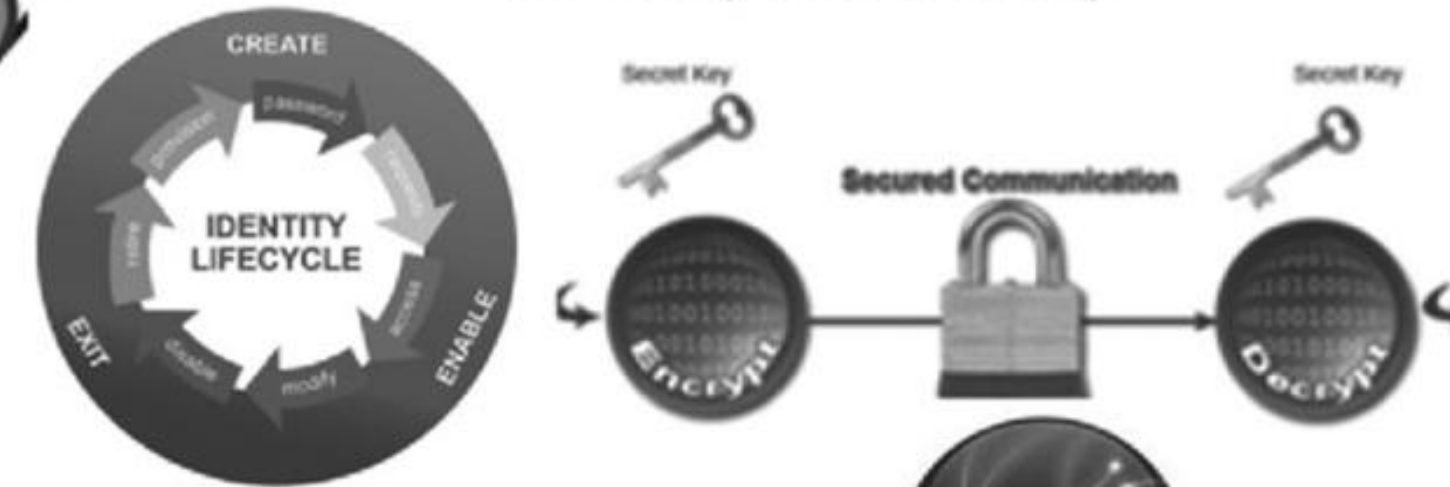


Keeping Humans in the Loop



· Human factor (regulate, improve)

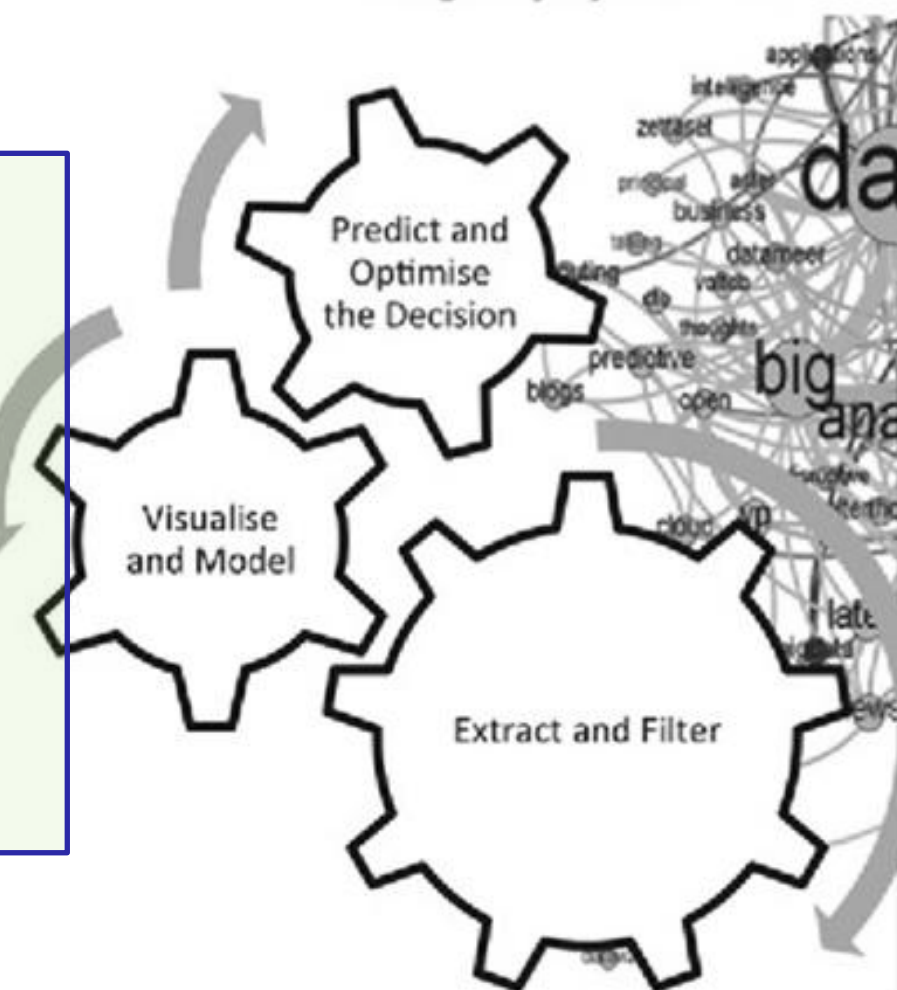
Security and Privacy



· Protection

· “Smartness”

· Big data
· Analytics



Connecting Objects and Humans



Data Analytics and Decision Making



Cloud Services



· Connectivity (protocols, etc,)

Outline

20

Course Logistics

Introduction

IoT design perspective

Basic Circuits

Conclusions

IoT Architecture (as a service)

21

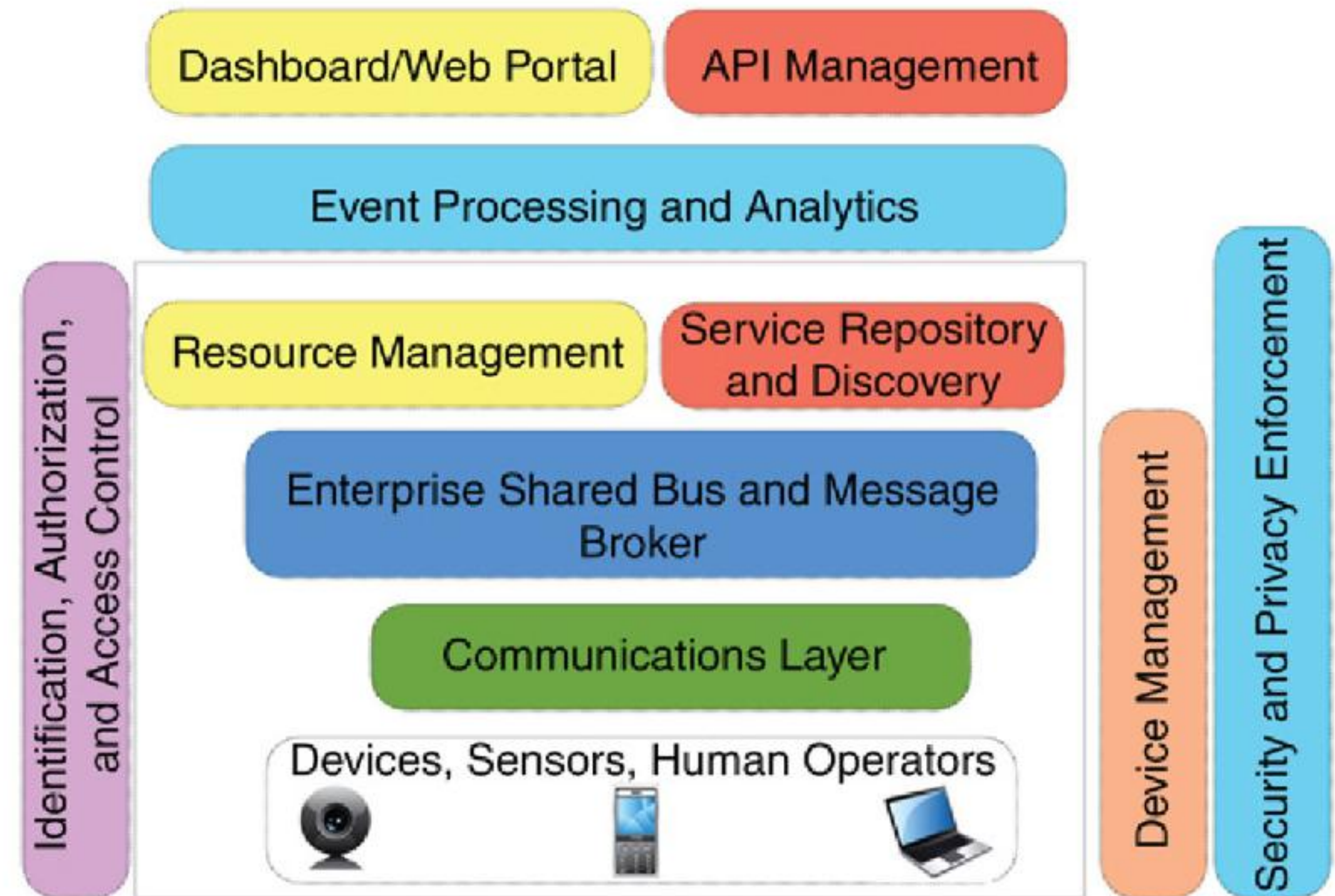
. Service Oriented Architecture (SOA):

1. **Sensing layer** is integrated with available hardware objects to sense the status of things
2. **Network layer** is the infrastructure to support over wireless or wired connections among things
3. **Service layer** is to create and manage services required by users or applications
4. **Interfaces layer** consists of the interaction methods with users or applications

Reference IoT implementation

22

- Executable program that complies with the I/O behavior of the standard.
 - May be written in a variety of language.
- Standards for communication, sensing, privacy, etc.



Platform design

23

- Required hardware to execute the IoT service.
- Turn system requirements and software models into detailed requirements.
 - Use profiling and analysis tools to measure existing executable specifications.
- Explore the design space manually or automatically.
- Develop hardware abstraction layers and other software.

Embedded systems for IoT

24

- **Platform-based (hardware) design**

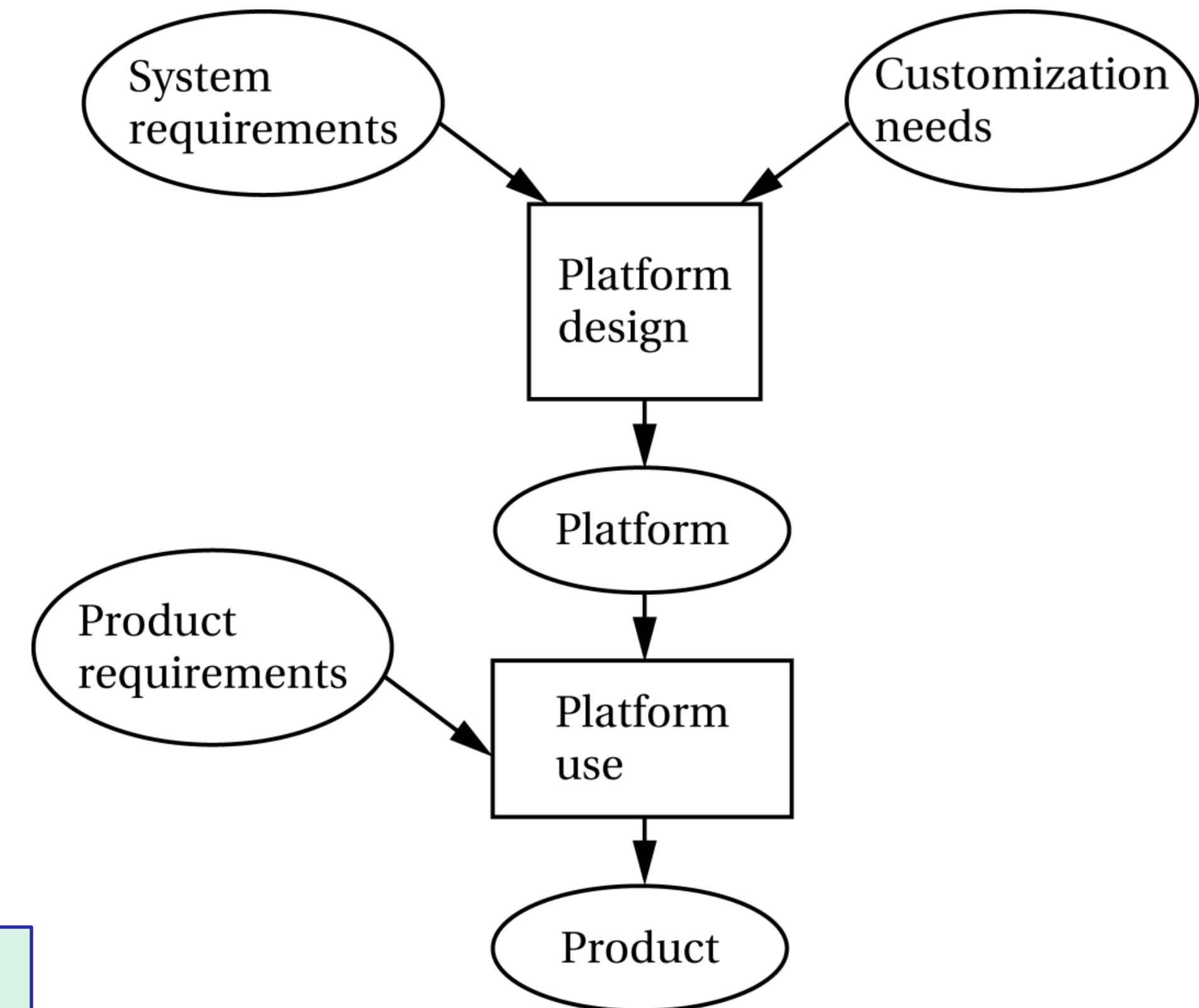
- Platform includes hardware, supporting software.

- **Two stage process:**

1. Design the platform.
2. Use the platform.

- Platform can be reused to host many different systems.

Smart sensing is the first requirement.



Outline

25

Course Logistics

Introduction

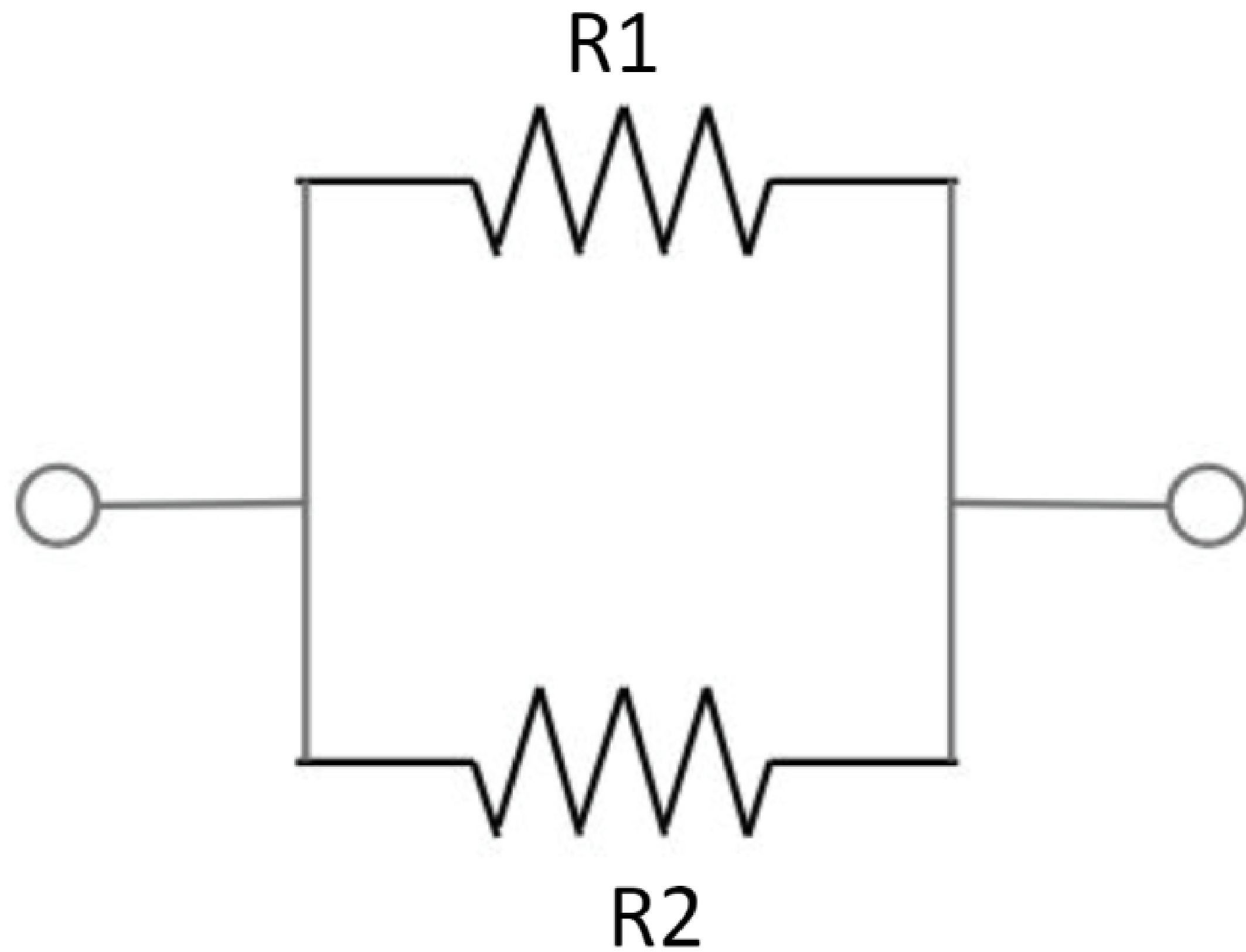
IoT design perspective

Basic Circuits

Conclusions

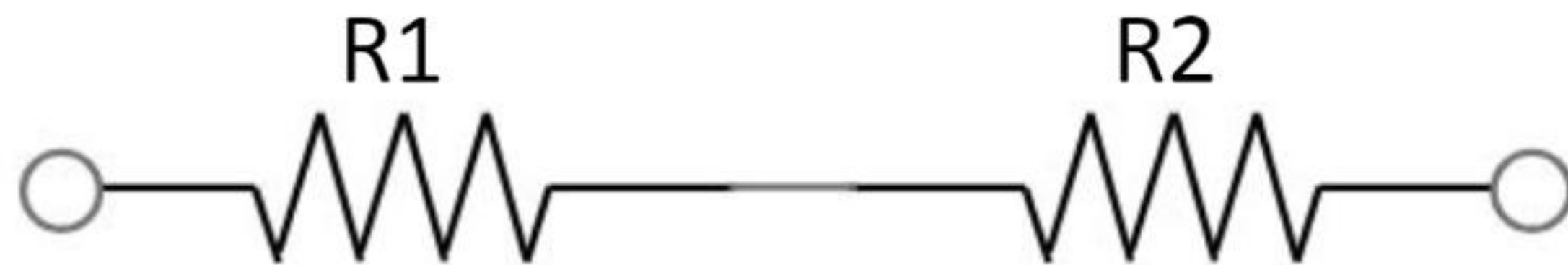
Resistencias en Paralelo

26



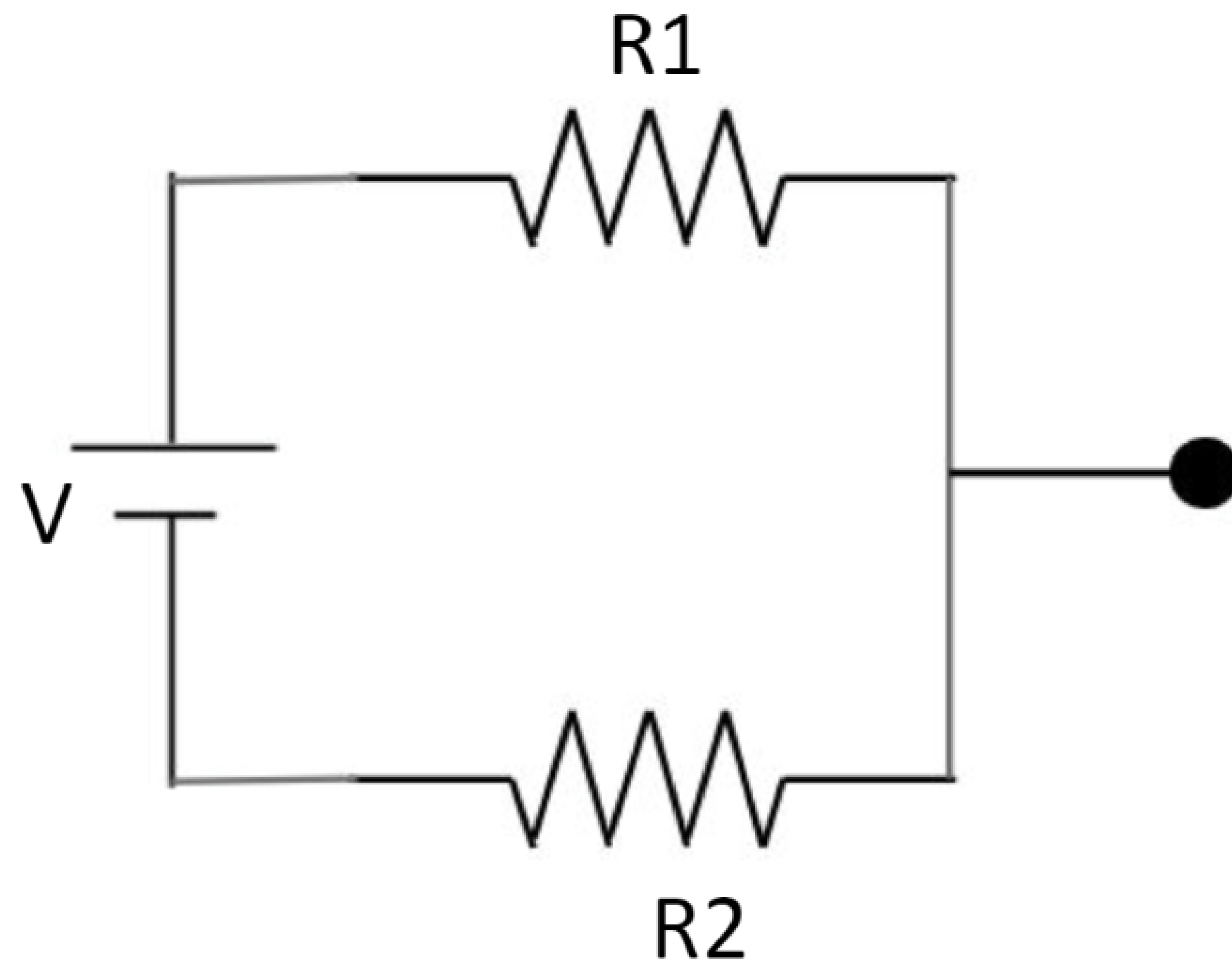
Resistencias en Serie

27



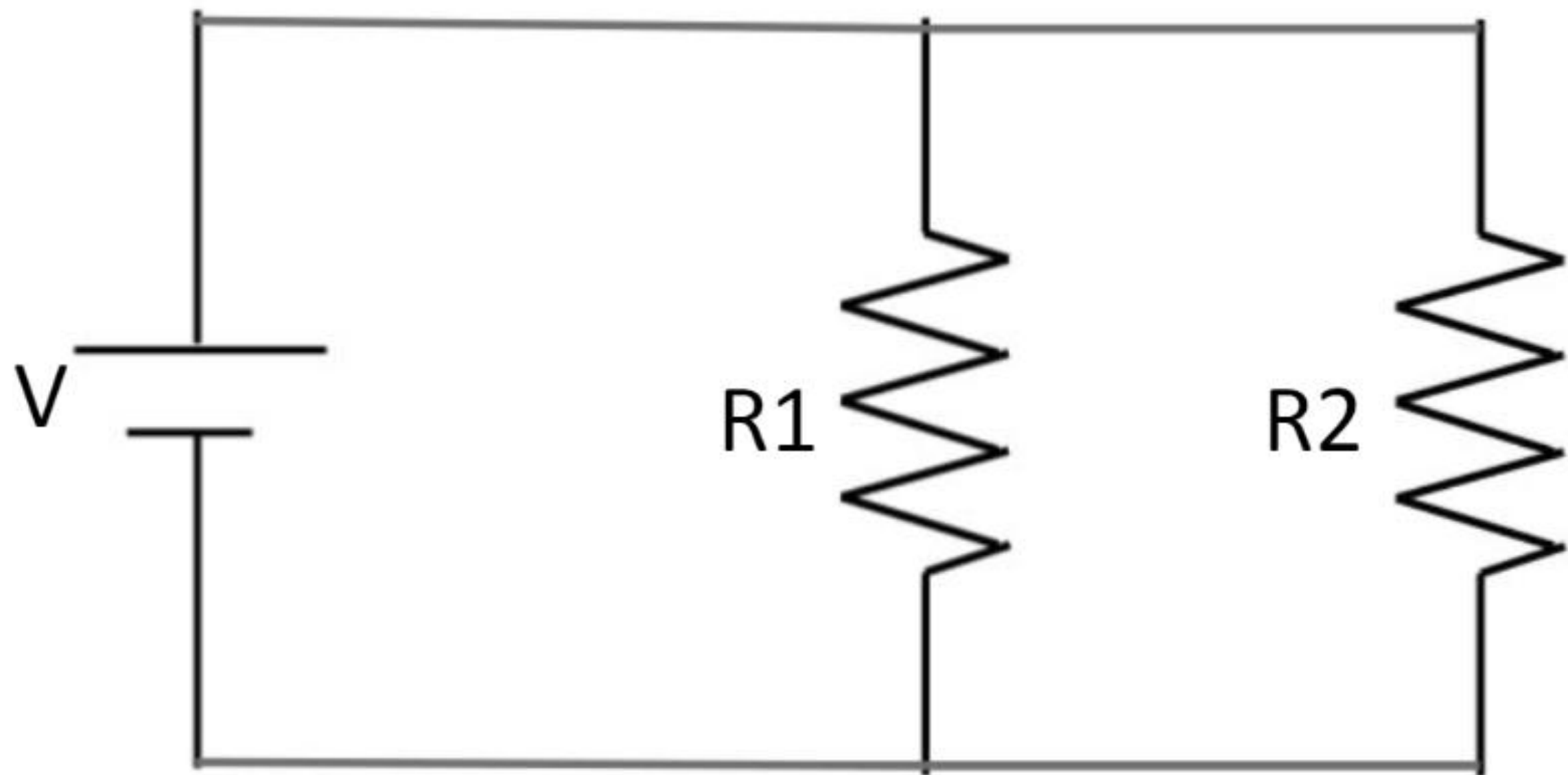
Divisor de Voltaje

28



Divisor de Corriente

29



Question 1

30

Tienes un sensor que funciona correctamente solo si recibe una señal de entrada entre **0 V y 3.3 V**.

Datos:

- La fuente de alimentación que tienes entrega **5 V**.
- Dispones de resistores con valores: **10 k Ω , 20 k Ω , 30 k Ω , 40 k Ω , 50 k Ω** .

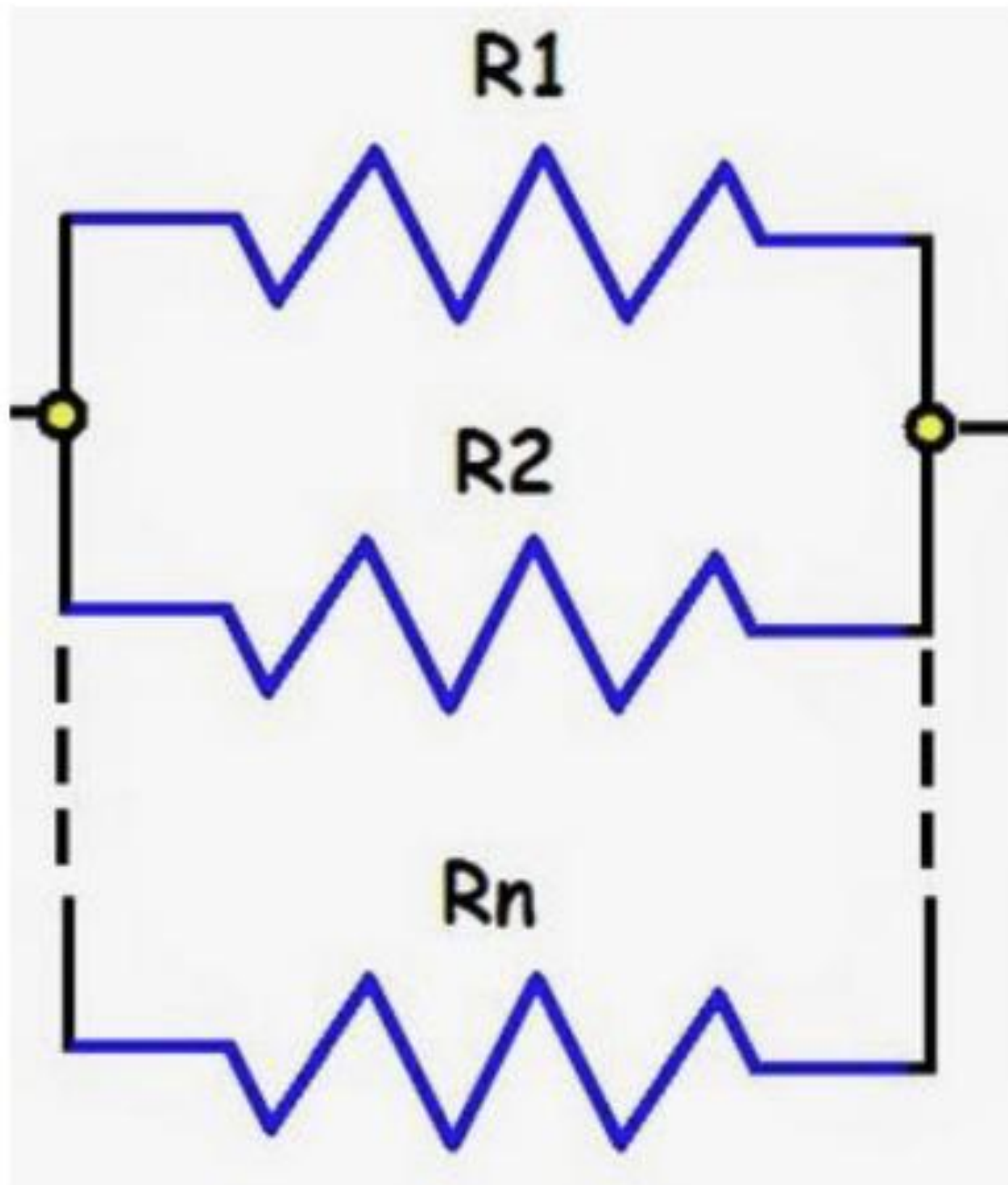
¿Cómo armarías un divisor de voltaje con dos resistores de los valores disponibles para que la salida hacia el sensor no supere los 3.3 V cuando la fuente entrega 5 V?

- Explica qué resistores elegirías.
- Calcula el voltaje de salida del divisor.
- Justifica por qué esa combinación es correcta para proteger al sensor.

Question 2

31

- ¿Cuál sería la fórmula general para una serie de n resistencias en paralelo?
Muestre cómo obtuvo la fórmula general.



Outline

32

Course Logistics

Introduction

IoT design perspective

Embedded systems for IoT

Conclusions

Summary

- We introduce our course: evaluation and objectives.
- We review the IoT design perspective:
 - Infrastructure and architectural
- We highlighted the relevance of the hardware platform for deployment of the IoT system.

1. Introduction

Internet of Things